Exercises

Lab 4

Exercises

i $a_{ij} = \frac{1}{(i+j-1)}$

$$H = \begin{bmatrix} 1 & 1/2 & 1/3 & 1/4 \\ 1/2 & 1/3 & 1/4 & 1/5 \\ 1/3 & 1/4 & 1/5 & 1/6 \\ 1/4 & 1/5 & 1/6 & 1/7 \end{bmatrix} \begin{array}{l} = 25/12 = 2.0833 \\ = 77/60 \\ = 57/60 \\ = \ldots \end{array}$$

$\|H\|_\infty = 2.0833$

$\kappa(H) = \|H\| \cdot \|H^{-1}\| = 2.0833 \cdot 13620 = 28,375$

Digits lost $= \log_{10}(28,375) = 4.45 \approx 4.5$ digits lost

$$\text{scaled} = \begin{bmatrix} 1 & 0.5 & 0.333 & 0.25 \\ 1 & 0.667 & 0.5 & 0.4 \\ 1 & 0.75 & 0.6 & 0.5 \\ 1 & 0.8 & 0.667 & 0.571 \end{bmatrix}$$

ii Norms

vector Norms

1- Norm $= \sum |x_i| : \sum_{i=1}^{n} 1 = n$

2-norm $= \sqrt{\sum |x_i|^2} : \sqrt{\sum_{i=1}^{n} 1^2} = \sqrt{n}$

infinity norm $= \max |x_i| : \max(1, 1, \ldots) = 1$

iii False because even if $\|A\|$ is small, $\|A^{-1}\|$ can still be large in $\kappa(A) = \|A\| \cdot \|A^{-1}\|$

iv True because the inverse of $\|A^{-1}\| = \|A\|$ and the opposite is also true.

Computer lab

i

```
A
     6    15    55
     5    55   225
    55   225   979

A_inv
    0.1192   -0.0855    0.0130
    0.2768    0.1054   -0.0398
   -0.0703   -0.0194    0.0094

Verification
    1.0000   -0.0000         0
   -0.0000    1.0000         0
         0         0    1.0000
```

ii

vector $v^{(k)}$ error

1 $[1.0085, 0.5043, 0.2596, 0.1348, 0.0724, 1.6412]^T$ 1

2 $[1.2589, 0.7593, 0.4470, 0.2547, 0.4005, 1.9552]^T$ 0.380

3 $[1.3847, 0.9158, 0.5878, 0.7441, 1.3497, 2.1798]^T$ 0.231267

1.64

1.2 - 33

1.4 - 16

1.6 - 24